Date.
Page No.

Section B

11) $a=b+c$
$d=b+c$
$e=a+d$
$f=b+c$
$g=e-f$

We assign numbers to unique expressions:-

Expression	Value Number
b	1
c	2
b+c	3

Now evaluate each statement:-
$a=b+c \rightarrow$ value $=3$
$d=b+c \rightarrow$ already exists $\rightarrow$ value $=3$
$e=a+d \rightarrow 3+3=$ new $\rightarrow 4$
$f=b+c \rightarrow$ already exists $\rightarrow$ value $=3$
$g=e-f \rightarrow 4-3=$ new $\rightarrow 5$

b+c appears three times so compute it once only.
Optimized code:-
$t=b+c$
$a=t$
$d=t$
$f=t$
$e=t+t$
$g=e-t$
- Introduced temporary variable t.
- Eliminated repeated computation of b+c.
- Reduced number of operations.